THEORETICAL MODEL

Reconnecting Dunhuang

The Library Cave and Its Manuscripts

STARTING PAGE	Wikipedia — “Dunhuang manuscripts”
MODEL SCOPE	15 entities · 16 normalized relations · 5 narrative layers
CORE TRAJECTORY	Together → Discovered → Dispersed → Institutionally Separated → Digitally Reconnected

Rationale and Modelling Decisions

1 Research Question and Project Framing

This project examines the historical trajectory of the Dunhuang Manuscripts from their original context in the Library Cave at Mogao Caves, through their discovery and subsequent dispersal, to their present-day preservation in different institutional collections and their digital reconnection through the International Dunhuang Project.

CENTRAL RESEARCH QUESTION

How can the historical movement, institutional dispersal, and digital reconnection of the Dunhuang Manuscripts be represented as a coherent information model?

Rather than treating the manuscripts only as a collection of cultural objects, the model focuses on the relationships that connect places, people, institutions, and digital infrastructures across different historical stages. It therefore represents the manuscripts through five interconnected layers: Original Context, Discovery, Dispersal, Collections, and Digital Reconnection.

Together, these layers describe a transition from physical and contextual unity to geographical and institutional separation, followed by a new form of connection in the digital environment. The theoretical model is designed to make this process explicit while preserving the distinction between historically documented relations and later forms of digital aggregation.

2 Entity Selection and Typing

The model is built around 15 selected entities that represent the principal places, people, institutions, documentary materials, and digital infrastructure involved in the trajectory of the Dunhuang Manuscripts. The selection is intentionally limited to entities that contribute directly to the historical and conceptual structure of the model, rather than attempting to represent every person, institution, or manuscript associated with Dunhuang.

For modelling purposes, these entities are organised into six theoretical types: Place, Person, Institution, Documentary Collection, Historical / Cultural Network, and Digital Heritage Project. These types provide a consistent semantic structure across the five layers while allowing entities with different roles to remain distinguishable.

TYPE DECISION

Heritage Site is treated as a subtype of Place rather than as a seventh independent theoretical type. Dunhuang, Mogao Caves, and Library Cave all belong to the broader Place category, while Mogao Caves and Library Cave can be identified more specifically as heritage sites.

The resulting entity set combines concrete historical actors and locations with institutional, documentary, network, and digital entities. This makes it possible to represent not only where the manuscripts were located and who participated in their discovery and dispersal, but also how they later became institutionally separated and digitally reconnected.

3 Relation Design and Normalization

The relationships between the selected entities were normalised to create a consistent structure across the model. The final model contains 16 relations, each representing one specific semantic connection between a subject and an object. Compound relationships were separated where necessary so that each edge expresses a single claim. For example, the role of the International Dunhuang Project is represented through separate relations for digitizes and provides_digital_access_to, rather than combining both functions into one relation.

NORMALIZATION RULE

All relations follow a consistent Subject → Relation → Object direction. Inverse formulations of the same historical fact are not added as separate relations.

For instance, Wang Yuanlu → sold_manuscripts_to → Aurel Stein is retained, while the inverse formulation Aurel Stein → acquired_from → Wang Yuanlu is not added separately. This prevents the same event from being represented twice simply from different perspectives.

Contextual information that qualifies a relation is also kept separate from the entity structure. Dates and scope descriptions are treated as annotations rather than additional entity nodes. The discovery date 25 June 1900, for example, qualifies the discovered relation, while the description “most remaining Chinese manuscripts; transfer to Beijing, 1910” defines the scope of Luo Zhenyu's facilitated_transfer_of relation. This keeps the entity set stable while preserving information needed to interpret individual relations.

KEY MODELLING DECISION — R13

The evidence supports two separate claims: Luo Zhenyu facilitated the transfer of most of the remaining Chinese manuscripts to Beijing in 1910, and these manuscripts are now held by the National Library of China. The model therefore keeps Luo Zhenyu → facilitated_transfer_of → Dunhuang Manuscripts separate from Dunhuang Manuscripts → held_by → National Library of China. No unsupported direct Luo Zhenyu → National Library of China relation is asserted.

Finally, relation labels such as part_of_network, sealed_in, held_by, and facilitated_transfer_of are used here as theoretical semantic labels. They describe the intended meaning of the relations at the Theoretical Model stage and should not yet be understood as formal properties from a specific ontology or metadata schema. Their alignment with existing ontologies and vocabularies is reserved for the subsequent Conceptual Model.

4 Five-Layer Narrative Architecture

The model is organised into five narrative layers that trace the changing context of the Dunhuang Manuscripts over time. These layers are not intended as independent datasets or formal classes. Instead, they provide a narrative structure through which the same central collection can be followed from its original physical context to its contemporary digital reconnection.

LAYER	STAGE	FUNCTION IN THE MODEL
I	**Original Context**	Establishes the historical and spatial setting. Dunhuang belongs to the Silk Road network; the Mogao Caves are located near Dunhuang; Library Cave (Cave 17) is located within the Mogao Caves; and the Dunhuang Manuscripts were sealed in the Library Cave.
II	**Discovery**	Introduces Wang Yuanlu and the rediscovery of the manuscripts on 25 June 1900. The date is retained as a temporal annotation rather than modelled as an independent entity.
III	**Dispersal**	Represents the movement of portions of the collection through Wang Yuanlu's transactions with Aurel Stein, Paul Pelliot, and Ōtani Kōzui. The model represents the process of geographical dispersal without attempting to reconstruct every individual manuscript movement.
IV	**Collections**	Shifts from historical dispersal to present institutional separation across the British Library, Bibliothèque nationale de France, National Library of China, and Tokyo National Museum. Luo Zhenyu's transfer role remains distinct from present-day custody.

LAYER	STAGE	FUNCTION IN THE MODEL
V	**Digital Reconnection**	Introduces the International Dunhuang Project. Through digitizes and provides_digital_access_to, IDP creates and exposes digital representations; the enriched relation virtually_brings_together expresses a broader reconnection of materials that remain physically dispersed.

CORE TRAJECTORY

Together → Discovered → Dispersed → Institutionally Separated → Digitally Reconnected

Digital reconnection is not presented as a reversal of historical dispersal. The physical separation of the manuscripts remains, while a new form of connection is created at the level of digital access and knowledge organisation.

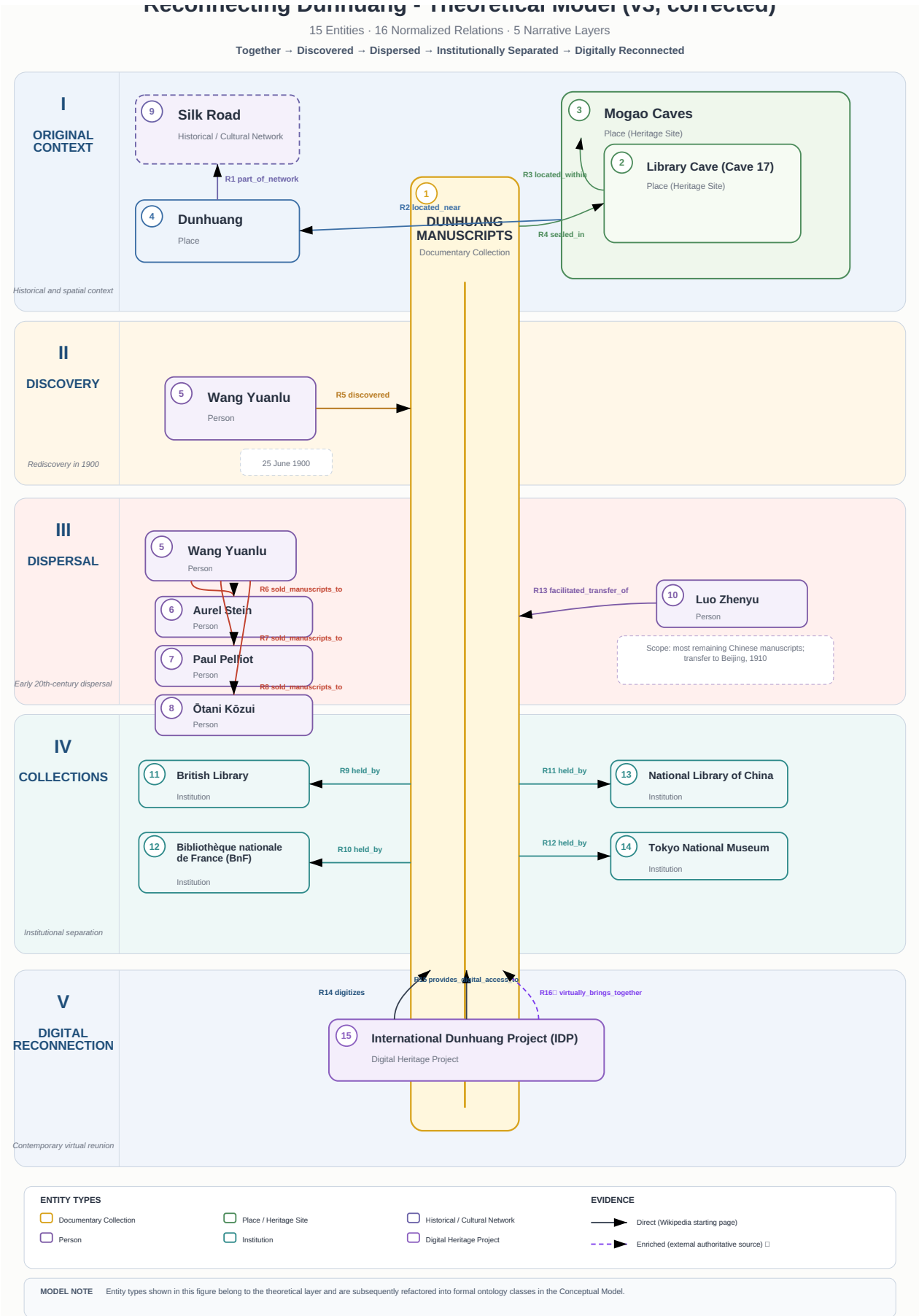


Figure 1. Theoretical model of the Dunhuang Manuscripts: five narrative layers, 15 entities, and 16 normalized relations.

5 Evidence and Source Strategy

The model distinguishes between Direct and Enriched relations in order to make the evidential basis of each claim explicit. The selected Wikipedia page, Dunhuang manuscripts, serves as the starting source for entity and relation identification. Relations that can be supported directly by this page are classified as Direct.

The model also allows limited enrichment when an important relation is not sufficiently supported by the starting page. In such cases, an additional authoritative source is used and the relation is explicitly marked as Enriched, rather than being presented as if it originated from the same source. In the current model, 15 of the 16 relations are Direct, while R16 — virtually_brings_together — is Enriched and supported by the International Dunhuang Project's institutional description of its mission.

EVIDENCE PRINCIPLE

Enrichment is permitted, but it must remain visible. The model records where an external source has been introduced instead of presenting every relation as if it had the same evidential origin.

This distinction makes the provenance of the model's claims more transparent and allows the model to extend beyond a single starting page without obscuring where additional interpretation or evidence has been introduced. The aim is not to collect as many relations as possible, but to retain a controlled set of relations that are relevant to the project's central narrative and can be supported by identifiable sources.

The complete entity-by-entity mapping, including identifiers, subject headings, classification information, verification status, and source links, is provided in Appendix B. Keeping this material in a separate appendix allows Figure 1 to remain readable while satisfying the requirement to connect relevant entities with authority-control and subject/classification information. These mappings also provide a controlled reference layer for the next stage, where the theoretical entities and relations will be compared with existing schemas, vocabularies, and ontologies in the Conceptual Model.

Authority mappings are treated as evidence-bearing references rather than as fields that must be filled at all costs. Where a reliable identifier could not be independently matched or verified, the mapping is recorded as NOT FOUND rather than assigning an uncertain identifier. This evidence-conscious approach preserves gaps transparently without treating incomplete authority coverage as a defect in the model. The National Library of China LC Name Authority File record (no2001051712) has been manually verified and is therefore recorded as CORROBORATED in Appendix B.

To connect the theoretical entities with established knowledge-organization systems, the model also records authority-control and subject/classification information for the 15 entities represented in Figure 1. Depending on the entity and the available record, these mappings include resources such as Wikidata, VIAF, GND, Library of Congress authority data and LCSH, BnF, NDL, FAST/WorldCat, UNESCO, and relevant institutional authority records. The purpose of these links is identification and disambiguation: they connect the model's theoretical entities to external controlled identifiers without yet replacing the model's provisional entity types or relation labels with a formal ontology.

6 Authority Control and Classification

7 Overall Modelling Rationale and Transition to the Conceptual Model

Overall, the theoretical model uses a controlled set of 15 entities and 16 normalized relations to represent the historical transformation of the Dunhuang Manuscripts from a collection preserved in a shared physical context to one dispersed across different institutions and subsequently reconnected through digital infrastructure. The five-layer structure provides the narrative framework for this transformation, while entity and relation normalization ensures that the same people, places, collections, and institutions are represented consistently throughout the model.

The model is intentionally selective rather than exhaustive. Its purpose is not to reproduce every entity or historical connection associated with the Dunhuang Manuscripts, but to identify the relationships necessary to explain the project's central idea: physical dispersal followed by digital reconnection. In this sense, the model provides a structured interpretation of the selected domain while remaining traceable to the evidence on which its relations are based.

At this stage, the entities and relations remain theoretical constructs expressed through provisional semantic types and relation labels. The next step is therefore to move from the Theoretical Model to the Conceptual Model, where these elements can be compared with and refactored through existing schemas, vocabularies, and ontologies. This separation keeps the present model focused on defining the domain and its relationships before decisions about

formal representation are introduced.

Appendix A. Normalized Entity–Relation Table

The table records the sixteen normalized relations used in the final theoretical model. Subject/object direction follows the arrowheads in the diagram; evidence level and contextual notes are retained as a checking layer.

#	SUBJECT	RELATION	OBJECT	EVIDENCE	NOTE
R1	Dunhuang	part_of_network	**Silk Road**	Direct	
R2	Mogao Caves	located_near	**Dunhuang**	Direct	*Corrected final relation; matches the diagram.*
R3	Library Cave (Cave 17)	located_within	**Mogao Caves**	Direct	
R4	Dunhuang Manuscripts	sealed_in	**Library Cave (Cave 17)**	Direct	
R5	Wang Yuanlu	discovered	**Dunhuang Manuscripts**	Direct	*25 June 1900*
R6	Wang Yuanlu	sold_manuscripts_to	**Aurel Stein**	Direct	
R7	Wang Yuanlu	sold_manuscripts_to	**Paul Pelliot**	Direct	
R8	Wang Yuanlu	sold_manuscripts_to	**Ōtani Kōzui**	Direct	
R9	Dunhuang Manuscripts	held_by	**British Library**	Direct	
R10	Dunhuang Manuscripts	held_by	**Bibliothèque nationale de France (BnF)**	Direct	
R11	Dunhuang Manuscripts	held_by	**National Library of China**	Direct	
R12	Dunhuang Manuscripts	held_by	**Tokyo National Museum**	Direct	
R13	Luo Zhenyu	facilitated_transfer_of	**Dunhuang Manuscripts**	Direct	*Scope: most remaining Chinese manuscripts; transfer to Beijing, 1910.*
R14	International Dunhuang Project	digitizes	**Dunhuang Manuscripts**	Direct	
R15	International Dunhuang Project	provides_digital_access_to	**Dunhuang Manuscripts**	Direct	
R16	International Dunhuang Project	virtually_brings_together	**Dunhuang Manuscripts**	Enriched	*Supported by IDP institutional description; not the starting Wikipedia page.*

APPENDIX B

15 Entities — Authority Control & Classification Mapping

Prepared for Reconnecting Dunhuang (Information Science and Cultural Heritage, a.y. 2025–2026). This table closes the Theoretical Model requirement to add “links to authority control and subjecting/classification information” before moving to the Conceptual Model stage.

METHOD AND CONFIDENCE LEGEND

Every ID below was checked against a live source before inclusion — nothing was pattern-guessed. Three confidence levels are used on purpose: a falsely complete-looking table would be worse than an honest, uneven one.

CONFIRMED the record page itself was opened and its content read (name / dates / description matched).

CORROBORATED backed by an independent live source (a search engine's index of the real page's title, or OCLC's FAST linked-data crosswalk) but the record page itself returned a 403/429 in this session. Very likely correct; a manual click-through before final submission is still worth doing.

NOT FOUND searched, no record located — a real, reportable result for several entities, not a gap in effort.

Several authority services, including viaf.org, id.loc.gov, and wikidata.org, rate-limited or blocked automated fetches during verification. As a result, several rows lean on CORROBORATED rather than CONFIRMED. Where OCLC's FAST crosswalk was used as the verification path for an LCSH/VIAF pair, that is noted inline.

Places and Documentary Collection

1 Dunhuang Manuscripts *Documentary Collection*

WIKIDATA	**CORROBORATED** **Q276864** https://www.wikidata.org/wiki/Q276864
VIAF	**CORROBORATED** **178621901** https://viaf.org/viaf/178621901
LCSH / LC AUTH.	**CORROBORATED** **“Dunhuang manuscripts”** https://id.loc.gov/authorities/names/n82066451
WORLDCAT	**CORROBORATED** **FAST fst01356877** https://id.worldcat.org/fast/1356877
NOTE	**NOT FOUND** — *Heading sits in the LC Name Authority File (MARC 130), not the Subjects file — still a valid controlled access point.*

4 Dunhuang *Place*

WIKIDATA	**CORROBORATED** **Q319114** https://www.wikidata.org/wiki/Q319114
VIAF	**CORROBORATED** **168042699** https://viaf.org/viaf/168042699

LCSH / LC AUTH.	CORROBORATED **"Dunhuang (China)"** https://id.loc.gov/authorities/names/nr96042661
WORLDCAT	CORROBORATED **FAST fst01278691** https://id.worldcat.org/fast/1278691
NOTE	NOT FOUND — *Also cross-linked to GeoNames 1529531.*

3 Mogao Caves *Place / Heritage Site*

WIKIDATA	CORROBORATED **Q43286** https://www.wikidata.org/wiki/Q43286
VIAF	NOT FOUND
LCSH / LC AUTH.	CORROBORATED **"Dunhuang Caves (China)" — "Mogao Caves" is a cross-reference term, not the authorized heading** https://id.loc.gov/authorities/subjects/sh86007577
WORLDCAT	CORROBORATED **FAST fst01240618 ("China—Dunhuang Caves")** https://id.worldcat.org/fast/1240618
NOTE	NOT FOUND — *Recommended citation form: "Dunhuang Caves (China) [also known as: Mogao Caves]". This specific heading could not be re-opened directly this session (403) — verify before quoting as fact.*

2 Library Cave (Cave 17) *Place / Heritage Site*

WIKIDATA	CORROBORATED **Q28097247 ("Dunhuang Cave 17")** https://www.wikidata.org/wiki/Q28097247
VIAF	NOT FOUND
LCSH / LC AUTH.	NOT FOUND — *no dedicated heading; catalogers fold this into the general "Dunhuang Caves" heading above*
WORLDCAT	NOT FOUND
NOTE	NOT FOUND — *Expected: too granular a sub-location for its own authority record.*

9 Silk Road *Historical / Cultural Network*

WIKIDATA	CORROBORATED **Q36288** https://www.wikidata.org/wiki/Q36288
VIAF	NOT FOUND
LCSH / LC AUTH.	CORROBORATED **"Silk Road" (variant: "Silk Route")** https://id.loc.gov/authorities/subjects/sh85122554
WORLDCAT	CORROBORATED **FAST fst01243730 ("Asia—Silk Road")** https://id.worldcat.org/fast/1243730
NOTE	NOT FOUND — *Also cross-linked to BnF: data.bnf.fr/ark:/12148/cb11970767h.*

PERSONS

5 Wang Yuanlu 王圓箓 *Person*

WIKIDATA	CORROBORATED **Q1202225** https://www.wikidata.org/wiki/Q1202225
VIAF	NOT FOUND
LCSH / LC AUTH.	NOT FOUND
WORLDCAT	NOT FOUND — *an unverified lead (entities.oclc.org, titled "Yuanlu Wang") surfaced but its content could not be confirmed — do not cite without checking*
NOTE	NOT FOUND — *Expected: thin/absent Western authority coverage for this figure, as anticipated by the assignment brief.*

6 Aurel Stein *Person*

WIKIDATA	CORROBORATED **Q313061** https://www.wikidata.org/wiki/Q313061
VIAF	CONFIRMED **88967071 — "Aurel Stein, Hungarian-British archaeologist, 1862–1943"** https://viaf.org/viaf/88967071/
LCSH / LC AUTH.	CORROBORATED **"Stein, Aurel, Sir, 1862-1943"** https://id.loc.gov/authorities/names/n82045000
WORLDCAT	CORROBORATED **id.oclc.org entity titled "Aurel Stein"** https://id.oclc.org/worldcat/entity/E39PBJcc8RpgyQVjD6fDkFB9Dq.html

7 Paul Pelliot *Person*

WIKIDATA	CORROBORATED **Q368821** https://www.wikidata.org/wiki/Q368821
VIAF	CONFIRMED **73858373 — "Pelliot, Paul; French scholar and explorer, 1878–1945" (ISNI 0000 0001 2102 3365)** https://viaf.org/viaf/73858373/
LCSH / LC AUTH.	CORROBORATED **"Pelliot, Paul, 1878-1945"** https://id.loc.gov/authorities/names/n50048941
WORLDCAT	NOT FOUND — *likely exists, not located*
NOTE	NOT FOUND — *A decoy VIAF ID (39412419 — actually Sir Percival David) was explicitly ruled out during research; only 73858373 is confirmed.*

8 Ōtani Kōzui 大谷光瑞 *Person*

WIKIDATA	CORROBORATED **Q386254** https://www.wikidata.org/wiki/Q386254
VIAF	CORROBORATED **113770129** https://viaf.org/viaf/113770129/

LCSH / LC AUTH.	CORROBORATED **“Ōtani, Kōzui, 1876-1948”** https://id.loc.gov/authorities/names/n85315990
WORLDCAT	NOT FOUND
NOTE	NOT FOUND — *Name-collision risk: do not confuse with Q7139850 (Ōtani Kazuko) or Q3594055 (Ōtani Kōshō), relatives who surface in the same searches.*

10 Luo Zhenyu 罗振玉 *Person*

WIKIDATA	CORROBORATED **Q716405** https://www.wikidata.org/wiki/Q716405
VIAF	CONFIRMED **16093476 — name forms “Luo Zhenyu” / “羅振玉” / “罗振玉”, 1866–1940, “Chinese writer, collector, and calligrapher”** https://viaf.org/viaf/16093476/
LCSH / LC AUTH.	CORROBORATED **“Luo, Zhenyu, 1866-1940” (variant: “贞松老人, 1866-1940”)** https://id.loc.gov/authorities/names/n81076827
WORLDCAT	NOT FOUND
NOTE	NOT FOUND — *Solid VIAF/LC coverage despite the WorldCat gap — a useful counter-example to “Chinese figures = no Western authority records.”*

INSTITUTIONS AND DIGITAL HERITAGE PROJECT

11 British Library *Institution*

WIKIDATA	CORROBORATED **Q23308** https://www.wikidata.org/wiki/Q23308
VIAF	NOT FOUND — *a candidate (180191787) surfaced but was tied to unrelated records (e.g. the Sherborne Missal) on spot-check — explicitly ruled out, not cited*
LCSH / LC AUTH.	CORROBORATED **“British Library; 大英图书馆”** https://id.loc.gov/authorities/names/n81139951
WORLDCAT	NOT FOUND — *WorldCat Identities is discontinued; the old URL 404s*
NOTE	NOT FOUND — *Don't confuse with BL sub-unit records (no92012745 “National Bibliographic Service”; nr87000127 “Document Supply Centre”).*

12 Bibliothèque nationale de France (BnF) *Institution*

WIKIDATA	CORROBORATED **Q193563** https://www.wikidata.org/wiki/Q193563
VIAF	NOT FOUND
LCSH / LC AUTH.	CORROBORATED **“Bibliothèque nationale de France”** https://id.loc.gov/authorities/no95028191
WORLDCAT	NOT FOUND

NOTE	NOT FOUND — *Don't conflate with Q19938912 ("BnF authorities") or Q15222191 ("general catalog of BnF").*

13 National Library of China *Institution*

WIKIDATA	CORROBORATED **Q732353** https://www.wikidata.org/wiki/Q732353
VIAF	NOT FOUND
LCSH / LC AUTH.	CORROBORATED — no2001051712; LC authority record manually verified.
WORLDCAT	NOT FOUND — *NLC contributes records as an OCLC cataloging partner, but no dedicated identity page was located*
NOTE	CORROBORATED — LC authority record manually verified; National Central Library, Taipei (Q618340, LC n50080371) is a different institution and should not be conflated with the National Library of China.

14 Tokyo National Museum *Institution*

WIKIDATA	CORROBORATED **Q653433** https://www.wikidata.org/wiki/Q653433
VIAF	NOT FOUND
LCSH / LC AUTH.	NOT FOUND
WORLDCAT	NOT FOUND
NOTE	NOT FOUND — *Distinct from "National Museum of Modern Art, Tokyo" (Q1359908) and the administrative umbrella "National Museum" (Q11421232).*

15 International Dunhuang Project (IDP) *Digital Heritage Project*

WIKIDATA	CORROBORATED **Q3754962** https://www.wikidata.org/wiki/Q3754962
VIAF	NOT FOUND — *expected — a project/website, not a cataloging institution*
LCSH / LC AUTH.	NOT FOUND — *expected*
WORLDCAT	NOT FOUND — *expected*
NOTE	NOT FOUND — *Officially rebranded "International Dunhuang Programme"; current site idp.bl.uk uses "Programme" — check the live Wikidata label before citing, it may lag the rename.*

Coverage Summary

15 of 15 entities have a confirmed or corroborated Wikidata QID. 6 of 15 entities have a confirmed or corroborated VIAF record — 3 directly confirmed by opening the record itself (Aurel Stein, Paul Pelliot, and Luo Zhenyu) and 3 corroborated through independent cross-references (Dunhuang Manuscripts, Dunhuang, and Ōtani Kōzui). The remaining nine — Library Cave (Cave 17), Mogao Caves, Wang Yuanlu, Silk Road, British Library, Bibliothèque nationale de France (BnF), National Library of China, Tokyo National Museum, and International Dunhuang Project (IDP) — have no VIAF record confirmed in this review and are therefore recorded as NOT FOUND rather than being assigned an unverified identifier.

11 of 15 carry a confirmed or corroborated LCSH/LC subject or name heading that doubles as the "subjecting/classification information" requested by the guideline: Dunhuang Manuscripts, Dunhuang, Mogao/Dunhuang Caves, Silk Road, Aurel Stein, Paul Pelliot, Ōtani Kōzui, Luo Zhenyu, British Library, Bibliothèque nationale de France (BnF),

and National Library of China. The remaining four — Library Cave (Cave 17), Wang Yuanlu, Tokyo National Museum, and International Dunhuang Project (IDP) — have no LCSH/LC authority record confirmed in this review, each for a documented and explainable reason noted in its entry above.